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Homology leakage selectively changes which model wins on genomic sequence-classification benchmarks

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Abstract

Motivation: Benchmarks for DNA sequence classification underpin method comparison, but standard train/test splits rarely control for sequence homology. When near-identical sequences span the split, models can score well by memorizing near-duplicates rather than generalizing, and the consequences for method comparison have not been systematically characterized.
Results: We audit 7 binary datasets from the Genomic Benchmarks suite, quantifying train/test homology by exact k-mer Jaccard similarity and re-evaluating four classifiers under a homology-aware re-split that confines near-duplicate clusters to one side. Two datasets are materially leaky (38–41% of test sequences have a training near-duplicate); correcting the split removes up to 16.4 points of accuracy (95% bootstrap CI $[15.8, 17.0]$), while the remaining datasets show no significant change (all clean-dataset accuracy deltas have 95% bootstrap confidence intervals including zero). Crucially, leakage does not merely inflate scores—it changes which method appears best: on both leaky datasets the apparent best model on the standard split—a random forest—is demoted after correction (to third of four and to last, respectively), whereas rankings on all five clean datasets are preserved. A homology-graded analysis indicates the mechanism is largely memorization of label-carrying near-duplicates—cleanly on one leaky dataset and partially on the other. We provide a per-dataset leakage report card and a drop-in homology-aware splitter.
Availability and implementation: Code and data are available at `https://github.com/rikhinkavuru/homology-leakage-genomic-benchmarks`; the audit is CPU-only and fully reproducible.

Keywords data leakage, sequence homology, benchmark evaluation, genomics, model ranking

1. Introduction

Machine learning for DNA sequence classification is now routine, and benchmark suites have become the default proving grounds for new methods. The Genomic Benchmarks collection (Grešová et al., 2023) is widely used precisely because it is small, accessible, and quick to run, making it a natural first stop when comparing models. The scientific value of any such suite, however, rests entirely on one assumption: that its train/test split measures generalization rather than recall of the training data.

Sequence homology threatens this assumption. We use *homology leakage* to mean the situation in which sequences related by shared ancestry or near-duplication appear on both sides of a split. A model can then succeed on the test set by recognizing memorized near-copies of training examples rather than by learning generalizable signal, so reported accuracy overstates true generalization. This failure mode is increasingly documented across biological machine learning, from regulatory-genomics regression to protein-interaction prediction and virtual screening (Section 2).

What remains underexamined is the structure and the consequences of this leakage within a single, widely used suite. Prior work documents inflated scores within individual leaky domains, but three questions are open: (i) whether leakage is uniform across a suite or concentrated in specific datasets; (ii) whether it changes *which* method appears best, not merely the absolute scores; and (iii) whether a controlled, within-study contrast against clean datasets can isolate leakage as the cause rather than a confound.

We address these questions for the Genomic Benchmarks binary classification suite. Our contributions are:

- a homology audit of the binary suite with a matched clean-versus-leaky control, showing that leakage is dataset-specific rather than universal;
- evidence that leakage *selectively changes which model appears best* on leaky datasets while leaving rankings on clean datasets intact;
- a homology-graded analysis indicating the mechanism is memorization of label-carrying near-duplicates, demonstrated cleanly on one dataset and partially on another; and
- a practical per-dataset report card and an open, drop-in homology-aware splitter.

The entire study is CPU-only, deterministic, and reproducible, with all seeds logged and no external services required.

2. Related work

Homology leakage is by now well established as a benchmarking hazard. hashFrag (Rafi et al., 2025) shows that homologous sequences spanning genomic train/test splits inflate the apparent performance of neural models and, in particular, that test performance scales with similarity to the training set; it provides tools to detect and partition such data. In protein–protein interaction prediction, Bushuiev et al. (2024) show that sequence- and metadata-based splits leave most test cases with near-duplicate counterparts, producing benchmarks that reward overfitting capacity rather than generalization. PINDER (Kovtun et al., 2024) similarly demonstrates that de-leaked test sets sharply reduce reported performance, including for AlphaFold-Multimer. In virtual screening, an audit of LIT-PCBA (Huang et al., 2025) shows that a trivial memorization baseline can match sophisticated methods once leakage is accounted for.

We differ from these studies in three ways. Each prior study examines a single leaky domain without a matched clean control; we audit a whole suite and find that leakage is dataset-specific, using a within-study leaky-versus-clean contrast to isolate it as the cause. Where prior work documents inflated absolute scores, we show that leakage changes which model appears best, quantified against clean datasets on which rankings are preserved. And we target the Genomic Benchmarks classification suite specifically—the community's default quick-evaluation benchmark—for which no such audit exists. Methodologically, we adapt the similarity-graded analysis used by hashFrag for regression to the classification setting and extend it to the consequence for method ranking.

3. Methods

3.1. Datasets

We study the seven binary datasets of the Genomic Benchmarks suite (Grešová et al., 2023), and additionally report the three-class *human_ensembl_regulatory* set separately. Per-dataset sizes, sequence lengths, and class balance are given in Table 1. Sizes range from 6,914 (*drosophila_enhancers_stark*) to 174,756 (*human_ocr_ensembl*) sequences; sequence lengths are fixed within some datasets (e.g. 251 bp for *human_nontata_promoters*, 200 bp for the demonstration sets) and variable in others; classes are balanced (0.50 positive fraction) except *human_nontata_promoters* (0.544).

3.2. Features and models

Each sequence is represented by overlapping k-mer counts for $k \in \{4, 6\}$ (4^k features), L2-normalized. We evaluate four classifiers spanning a capacity range: logistic regression (`C=1.0`, `max_iter=5000`), a linear support-vector machine (`LinearSVC`, `C=1.0`, `dual=False`, `max_iter=5000`), a random forest (150 trees), and gradient-boosted trees (`HistGradientBoosting`, library defaults). All models use a fixed random seed; performance is reported as accuracy, AUROC, and F_1.

3.3. Homology measurement

We measure homology by the exact Jaccard similarity of k-mer sets with $k = 8$, computed for all test–train pairs via sparse binary matrix products (no MinHash approximation). For each test sequence we record its maximum similarity to any training sequence, and define the leakage fraction as the proportion of test sequences whose maximum similarity exceeds a threshold; we report thresholds 0.5, 0.7, and 0.9. We detect near-duplicate leakage via exact k-mer Jaccard similarity; this captures near-identical sequences, a detectable and conservative subset of homology leakage in the broader evolutionary sense (diverged homologs with low k-mer overlap are not flagged and would, if present, only add undetected leakage).

3.4. Homology-aware re-split

To correct leakage we build a similarity graph in which sequences are connected when their k-mer Jaccard exceeds 0.7, take connected components as near-duplicate clusters, and assign whole clusters to either train or test. The assignment preserves the original train/test ratio and class balance and, by construction, guarantees zero residual cross-split similarity above the threshold (verified empirically). We confirm robustness to the clustering threshold (sweeping $0.5/0.7/0.9$) and to removal of the single largest component. Sequence-clustering tools such as CD-HIT (Li and Godzik, 2006) and MMseqs2 (Steinegger and Söding, 2017) are the established means of homology-aware partitioning in bioinformatics; our exact-Jaccard procedure is a lightweight, dependency-free alternative suited to this audit, and either could substitute in the splitter.

3.5. Controls

Three controls isolate and explain the effect. (i) A *random* re-split at matched size and class balance, ignoring clusters, serves as a negative control. (ii) A homology-graded analysis bins test sequences by their maximum similarity to training and reports per-bin accuracy. (iii) A label-concordance analysis records, for each near-duplicate test sequence, whether its nearest training neighbour shares its label.

3.6. Reproducibility

All experiments are deterministic with logged seeds, CPU-only, and free of external services. The homology-aware splitter is released as a standalone tool (`homology_split.py`, dependencies `numpy`/`scipy`). Reported 95% confidence intervals are percentile bootstrap intervals (the 2.5th and 97.5th percentiles of $1{,}000$ resamples of the per-test-example correctness vector; the resampling unit is the individual test sequence, models are not refit, and the random seed is fixed).

4. Results

4.1. Leakage is dataset-specific

Of the seven binary datasets, only two—*human_nontata_promoters* and *human_enhancers_ensembl*—exceed a 10% near-duplicate test fraction at Jaccard 0.7 (0.406 and 0.384 respectively); the remaining five are near zero (Table 2). The three-class *human_ensembl_regulatory* set is likewise clean (0.005). Leakage is therefore concentrated, not endemic to the suite.

4.2. Correcting the split lowers accuracy only on leaky datasets

Under the homology-aware re-split, accuracy drops sharply on the leaky datasets—by 16.4 points (95% bootstrap CI $[15.8, 17.0]$) for the random forest on *human_enhancers_ensembl* and 10.8 points (95% CI $[9.9, 11.8]$) on *human_nontata_promoters*—whereas clean-dataset deltas are not statistically significant (all five 95% bootstrap CIs include zero). Both leaky-dataset random-forest drops are significant (CIs exclude zero). The negative control confirms causation: a random re-split of the same data, at matched size and balance, leaves the leaky-dataset accuracies essentially unchanged (best-model deltas -0.001 and -0.002, with 95% CIs including zero). Only removing homology, not re-splitting per se, lowers the score (Fig. 1).

4.3. Leakage changes which model appears best

The consequence for method comparison is the central result. On both leaky datasets the apparent best model on the standard split—a random forest—is demoted after correction, falling from rank 1 to rank 3 on *human_nontata_promoters* and from rank 1 to last (rank 4) on *human_enhancers_ensembl*, whereas the ranking is preserved on all five clean datasets (Fig. 2). Summarized by rank correlation, the mean Kendall τ between the leaky-split and corrected rankings is -0.17 on leaky datasets versus $+0.67$ on clean ones; with only four models, however, τ is restricted to a coarse set of values ($0, \pm0.33, \pm0.67, \pm1$, and the -0.17 is the mean of -0.33 and 0), so we treat it as descriptive support for the concrete demotion above rather than as a primary statistic. Material rank changes occur on $2/2$ leaky datasets and $0/5$ clean datasets; the clean-dataset changes are swaps between statistically tied models (overlapping accuracy CIs)—every such swap is between models whose 95% bootstrap CIs overlap, including *drosophila_enhancers_stark*'s larger (≈ 0.019) swap, which is additionally unstable across re-split seeds.

4.4. The mechanism is memorization of label-carrying near-duplicates

Binning test sequences by similarity to training reveals the mechanism directly (Fig. 3). On *human_enhancers_ensembl*, the random forest scores 1.000 on near-duplicate test sequences (≥ 0.9 similarity) but only 0.770 on novel ones (< 0.5), a gap of $+0.230$, versus a gap of roughly 0.035 for the other models; tellingly, on the novel sequences the random forest (0.770) is *worse* than the linear SVM (0.782). The label-concordance analysis explains why memorization pays: at ≥ 0.9 similarity, 99.95% (nontata) and 99.99% (ensembl) of near-duplicate test sequences share the label of their nearest training neighbour, so a model that recognizes a near-copy is correct essentially for free. On *human_enhancers_ensembl* the inflation is confined to the high-capacity model: the random-forest drop is significant (16.4 points, 95% CI $[15.8, 17.0]$, excludes zero) while the logistic-regression drop is null (≈ 0.000, 95% CI $[-0.006, +0.007]$, includes zero), indicating that only the model able to memorize near-duplicates is inflated. Clean datasets have near-empty high-similarity bins, leaving nothing to memorize.

4.5. Honest scope of the mechanism

The two leaky datasets differ in how cleanly the memorization account applies. On *human_enhancers_ensembl*, two independent evaluation routes agree: the corrected split and a novel-sequences-only ranking both dethrone the random forest. On *human_nontata_promoters*, these routes diverge ($\tau = -0.67$ between the corrected and novel-only rankings): the random forest is demoted under the corrected split but remains the best model on truly novel sequences (accuracy 0.879). Its advantage there is therefore partly genuine generalization, not pure test-side memorization. The inflation and ranking-change findings hold for both datasets; the mechanistic claim is clean for *human_enhancers_ensembl* and partial for *human_nontata_promoters*. This implies a scope limit for homology-aware splitting: it reliably detects and removes near-duplicate leakage, but because cluster removal can also discard legitimately learnable signal, the corrected ranking is not guaranteed to recover the true generalization ranking—as *human_nontata_promoters* shows, where the random forest retains a genuine advantage on novel sequences.

5. Discussion

Reported performance on some popular genomic benchmarks overstates generalization, and—more consequentially—can mislead about which method is best. Anyone using these suites to rank models risks selecting a method for its capacity to memorize near-duplicates rather than to generalize.

Because the effect is dataset-specific, the contribution is constructive rather than purely cautionary: not "these benchmarks are broken" but "here is which ones to trust, and a tool to certify any new dataset." The clean datasets passing the same checks act as a positive control that makes the leaky-dataset findings credible. We package this as a per-dataset report card and a drop-in splitter.

A practical caution concerns scale. Subsampling can mask leakage: *human_enhancers_ensembl* appears almost clean at a 20,000-sequence subsample (leakage fraction 0.056 at Jaccard 0.7) yet is heavily leaky at full scale (0.384), because subsampling breaks up near-duplicate pairs. Homology audits must therefore be run at full dataset scale.

Our study has limits. We use classical models rather than deep networks; we expect the mechanism to generalize to other model classes, though we demonstrate it only for the classical models studied here, and the graded analysis supports it; we use k-mer features and exact k-mer Jaccard as the similarity proxy (validated against the re-split guarantee); and we examine a single benchmark suite. Natural extensions are deep models, additional suites, and alignment-based similarity measures.

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Table 1. Datasets in the Genomic Benchmarks binary suite (and the optional three-class set). Lengths are the median (with min–max range where variable); class balance is the positive-class fraction (binary).

Dataset	n (full)	Length (bp)	Class balance	Test frac.
human_nontata_promoters	36,131	251	0.544	0.25
human_enhancers_ensembl	154,842	269 (2–573)	0.500	0.20
human_ocr_ensembl	174,756	315 (71–593)	0.500	0.20
human_enhancers_cohn	27,791	500	0.500	0.25
demo_coding_vs_intergenomic_seqs	100,000	200	0.500	0.25
demo_human_or_worm	100,000	200	0.500	0.25
drosophila_enhancers_stark	6,914	2142 (236–3237)	0.500	0.25
human_ensembl_regulatory (3-class)	289,061	401 (89–802)	3 classes	0.20

Table 2. Leakage report card. Verdict is LEAKY if the full-scale near-duplicate test fraction exceeds 0.1 at Jaccard 0.7. "Acc. lost" is the best-model accuracy change under the homology-aware re-split (original − corrected; positive = accuracy lost under correction); "RF rank" is the random-forest rank before/after correction.

Dataset	n (full)	leak@0.7	leak@0.9	Verdict	Best model	Acc. lost	Rank change?	RF rank
human_nontata_promoters	36,131	0.406	0.225	**LEAKY**	RF (k=6)	$+0.108$	yes	$1 \to 3$
human_enhancers_ensembl	154,842	0.384	0.380	**LEAKY**	RF (k=6)	$+0.164$	yes	$1 \to 4$
demo_coding_vs_intergenomic_seqs	100,000	0.078	0.024	clean	LR (k=6)	-0.001	no	$4 \to 4$
human_ocr_ensembl	174,756	0.010	0.001	clean	LR (k=6)	$+0.004$	no	$4 \to 4$
demo_human_or_worm	100,000	0.012	0.003	clean	LR (k=6)	-0.004	no	$4 \to 4$
drosophila_enhancers_stark	6,914	0.016	0.006	clean	LR (k=6)	$+0.012$	no	$2 \to 3$
human_enhancers_cohn	27,791	0.001	0.000	clean	LR (k=6)	-0.004	no	$4 \to 4$
human_ensembl_regulatory (3-class)	289,061	0.005	0.001	clean	RF (k=4)	-0.010	n/a	n/a

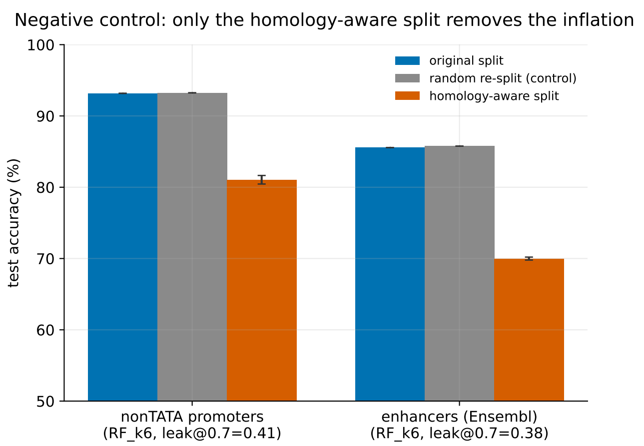

Figure 1 Original split versus random re-split (negative control) versus homology-aware re-split, best model per dataset. Only the homology-aware split lowers accuracy, and only on leaky datasets. The y-axis is truncated at 50% (not 0%) to make the homology-aware accuracy drop visible; bars span best-model accuracies of roughly 70–93%.

Conflict of Interest

None declared.

Data Availability

All code required to reproduce the analyses, along with the leakage report card, figures, and the homology-aware splitter tool, is available at `https://github.com/rikhinkavuru/homology-leakage-genomic-benchmarks`. The Genomic Benchmarks datasets analysed here are publicly available and were obtained via the genomic-benchmarks package (Grešová et al., 2023); they are not redistributed in our repository.

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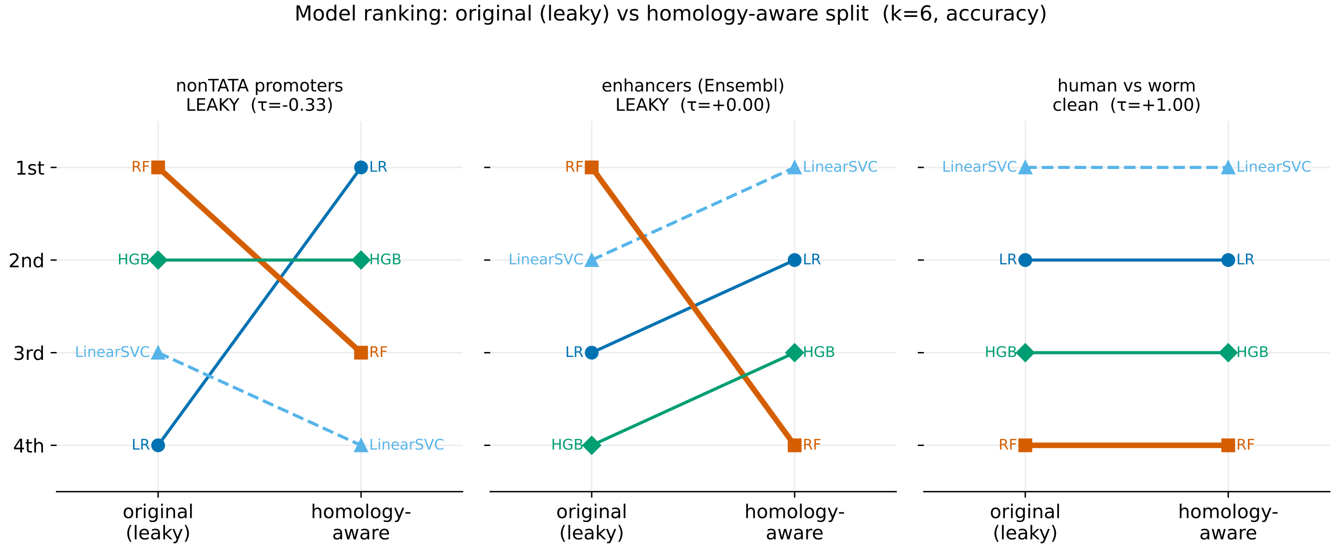

Figure 2 Model rankings on the original (leaky) split versus the homology-aware split (k=6, accuracy). The random forest is dethroned on the leaky datasets (left, middle) but rankings are stable on a clean control (right). Per-panel τ is the rank correlation for that dataset; on enhancers (Ensembl) the random forest's move to last flips only its own pairs, giving $\tau \approx 0$, while the -0.17 in §4.3 is the mean across both leaky datasets.

10.1038/nbt.3988.

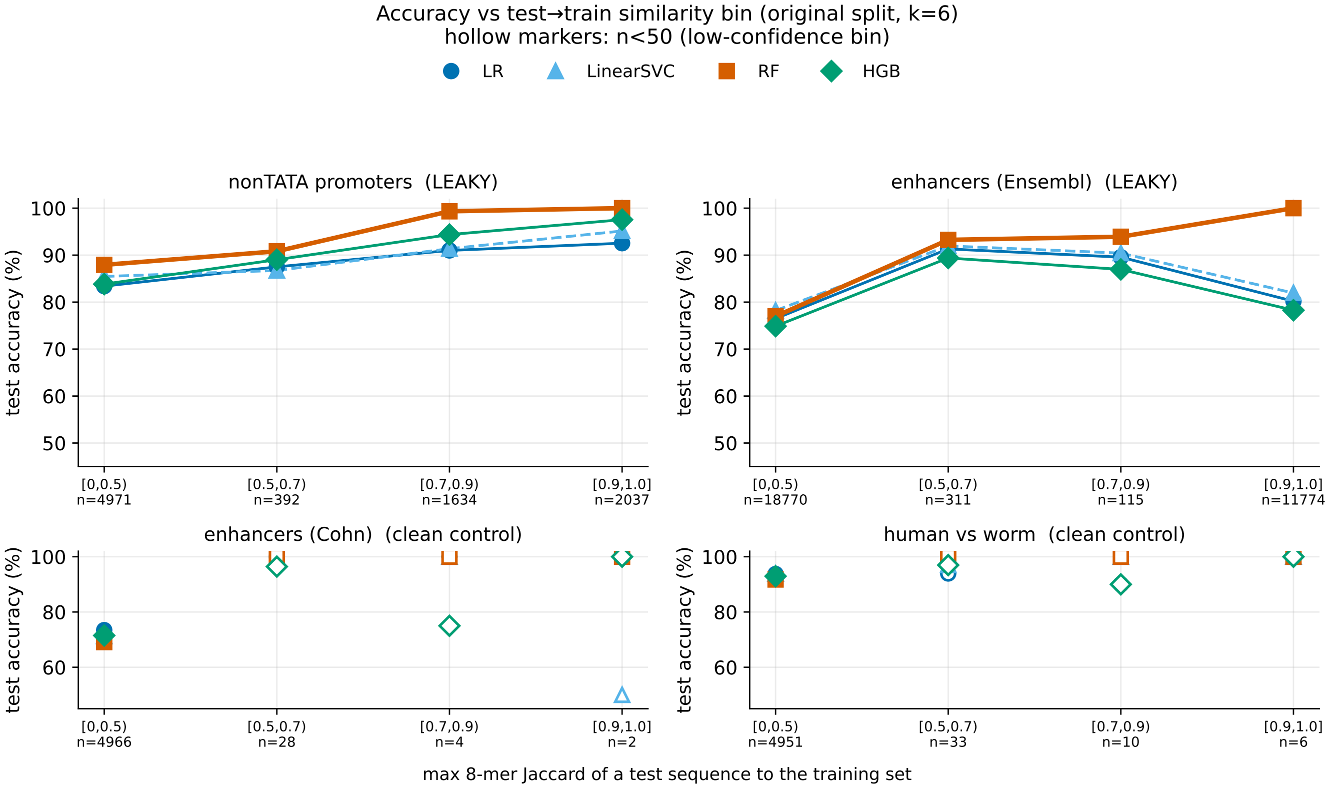

Figure 3 Accuracy as a function of the test-to-train similarity bin (original split, k=6). On leaky datasets the random forest is near-perfect on high-similarity (near-duplicate) test sequences and falls on dissimilar ones; clean controls have near-empty high-similarity bins (hollow markers: $n < 50$).